Integral formulas for quaternion valued forms, and their applications to SU(2) Yang-Mills theory

Edward B. Baker III

University of Maryland, College Park

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Abstract

We derive a formula relating the integral of a holomorphic function multiplied by an anti-holomorphic function to a corresponding double integral over the boundary. This result is then generalized to multiple complex variables, and to the quaternions. The quaternionic version requires new mathematics that is naturally related to SU(2) Yang-Mills theory, and this relationship is explored further. In particular, we derive an integral representation for the Lagrangian of Yang-Mills theory that promises to be useful for the quantum theory. This application, and generalizations to different gauge groups, could provide some insight into non-perturbative gauge theory, and seems related to both holography and dual-conformal symmetry.

1 Introduction

The quaternions are an extension of the complex numbers, satisfying the multiplication rules

$$\mathrm{e}_i^2 = -1, \quad [\,\mathrm{e}_i\,,\mathrm{e}_j\,] = 2\,\epsilon_{ijk}\,\mathrm{e}_k \quad \forall\,\mathrm{i,j,k} \in \{1,2,3\}.$$

Hamilton developed the quaternions in large part to describe three dimensional rotations. He realized that in order to do so naturally, it was necessary to include a fourth dimension. It has since been realized that four dimensions is the natural setting for relativistic physics, and much of the mathematics used in general relativity and four dimensional quantum field theory can be formulated with quaternions [1][2]. In addition, quaternions are a useful tool in the study of SU(2) Yang Mills theory, and were important for the classification of instantons (for some gauge groups) [3]. There has also been recent interest in hypercomplex geometry (and its variants) in both the mathematics and physics communities [4][5], in addition to the related twistor programme initiated by Roger Penrose [6][7].

There have been a number of attempts to extend the notion of complex analyticity to the quaternions, with some success. The earliest attempt to do this was by Fueter [8][9], related to work by Hamilton, which defines a quaternion regular function by

$$\partial_{\bar{q}} f \equiv \frac{1}{2}(\partial_0 + i\partial_1 + j\partial_2 + k\partial_3)f = 0,$$

which is called the Cauchy-Riemann-Fueter (CRF) equation. This approach has been extended in a number of directions, notably by Dominic Widdows in his thesis [10], related to work by Dominic Joyce [11]. In that work, the CRF equation is extended to an arbitrary hypercomplex manifold, and interpreted as a projector onto an invariant subspace of the Lie algebra defined by the generators

$$\mathcal{I}(\omega) = I(\omega) - \omega i,\ \mathcal{J}(\omega) = J(\omega) - \omega j,\ \mathcal{K}(\omega) = K(\omega) - \omega k,$$

where I, J, and K are the three complex structures associated to the hypercomplex manifold, satisfying the relation $IJ = K$. A different approach was also developed by Cullen [12], which is more closely related to the expansion of a complex analytic function in its power series, and has also been developed extensively [13].

2 Complex integral formulas

To begin, we will consider the integral

$$I = \int_\Omega f(z)g(\bar{z})\, dz \wedge d\bar{z},$$

defined on a closed, compact domain Ω. We first use the Cauchy integral formula for both f(z) and $g(\bar{z})$ to write

$$I = -\frac{1}{4\pi^2}\int_\Omega \left(\oint_{\partial\Omega} \frac{f(z_1)}{z_1 - z} dz_1 \oint_{\partial\Omega} \frac{g(\bar{z}_2)}{\bar{z}_2 - \bar{z}} d\bar{z}_2\right) dz \wedge d\bar{z}.$$

Assuming this integral does not diverge, we can interchange the loop integrals and the integral over z to obtain

$$I = -\frac{1}{4\pi^2}\oint_{\partial\Omega}\oint_{\partial\Omega} f(z_1)K(z_1, \bar{z}_2)g(\bar{z}_2)dz_1 d\bar{z}_2, \tag{2.1}$$

where we have defined $K(z_1, \bar{z}_2) = \int_\Omega (z - z_1)^{-1}(\bar{z} - \bar{z}_2)^{-1}\, dz \wedge d\bar{z}$. Using the distributional equation $\frac{\partial}{\partial \bar{z}}\frac{1}{z} = \pi\delta^2(z)$, we find that $K(z_1, \bar{z}_2)$ satisfies the differential equation

$$\frac{\partial}{\partial \bar{z}_1} K(z_1, \bar{z}_2) = \frac{\pi}{\bar{z}_1 - \bar{z}_2}, \tag{2.2}$$

with a similar equation for z_2. We therefore find that

$$K(z_1, \bar{z}_2) = \pi \log(|z_1 - z_2|)$$

which is a Green's function for the Laplacian.

We can generalize this formula to higher complex dimension, and then to the quaternions, which will be physically relevant. For the case of n complex variables with $n > 1$, we define a holomorphic form $\omega \in \Omega^{(n,0)}$ and an anti-holomorphic form $\nu \in \Omega^{(0,n)}$, where $\Omega^{(p,q)}$ denotes the space of complex forms of type (p,q). We will derive a formula similar to (2.1) for the integral

$$I_n = \int_\Omega \omega \wedge \nu,$$

in some bounded region $\Omega \in \mathbb{C}^n$. Following the derivation in the complex case, we find an analogue of equation (2.1), with $K(z, w) \in \Omega_1^{(n-1,0)} \otimes \Omega_2^{(0,n-1)}$,

$$I_n = C_n \int_{\partial\Omega}\int_{\partial\Omega} \omega(z) \wedge K_n(z, w) \wedge \nu(w). \tag{2.3}$$

Here K(z, w) must satisfy an equation for the Dolbeault operator, similar to equation (2.2). This solution is related to the Bochner-Martinelli kernel, and we can use the Bochner-Martinelli formula to show,

$$K_n(z,w) = C'_n \sum_{j=1}^{n} |z-w|^{2-2n} \bigwedge_{l\neq j} \mathrm{d}z_l \otimes \bigwedge_{m\neq j} \mathrm{d}\bar{w}_m.$$

We can also derive this result in the reverse direction. Starting with equation (2.3), I_n can be rewritten using Stoke's theorem as

$$I_n = C_n \int_\Omega \int_\Omega (\partial_z + \bar{\partial}_z)(\partial_w + \bar{\partial}_w)(\omega(z) \wedge K_n(z,w) \wedge \nu(w)).$$

Because $\omega \in \Omega^{(n,0)}$ and $\bar{\partial}_z \omega = 0$, we have

$$(\partial_z + \bar{\partial}_z)\left(\omega(z) K_n(z,w)\right) = (-1)^n \omega \wedge \bar{\partial}_z K_n(z,w),$$

and a conjugate result for ν. From this, we see that if K(z,w) satisfies the equation

$$C_n\, \bar{\partial}_z \partial_w K_n(z,w) = \delta^{2n}(z-w) \bigwedge_l \mathrm{d}z_l \otimes \bigwedge_m \mathrm{d}\bar{w}_m + \dots, \tag{2.4}$$

then the result follows immediately, assuming that the extra terms are not of type $\Omega_1^{(n,0)} \otimes \Omega_2^{(0,n)}$. These cross terms will be considered more explicitly in the quaternionic case. This also explains why $K_n(z,w)$ contains a Green's function for the Laplacian, because if we assume that K_n is only a function of $z' = z - w$, then the left side of equation (2.4) becomes $-C_n \Delta K_n(z-w)$.

3 Quaternion differentials and forms

The generalization to the quaternions is not straightforward, and requires new mathematics that is surprisingly powerful. It is also directly applicable to Euclidean SU(2) Yang-Mills theory in four-dimensions. As motivation for our approach, we consider the formula for the complex differential

$$\mathrm{d}f = \partial_z f \mathrm{d}z + \partial_{\bar{z}} f \mathrm{d}\bar{z}.$$

This formula immediately suggests the splitting between holomorphic and anti-holomorphic functions that is so important in the study of complex analysis. There are a few ways of extending this formula to the quaternions, but one that is both useful and suggestive is

$$\mathrm{d}f = \mathrm{d}\bar{q}\ \partial_{\bar{q}} f + \sum_i (\mathrm{e}_i \mathrm{d}\bar{q}\, \bar{\mathrm{e}_i})(\mathrm{e}_i \partial_{\bar{q}} \bar{\mathrm{e}_i}\, f). \tag{3.1}$$

From this formula, we see that the CRF operator appears naturally in the expression for the quaternion differential. We define the vector fields and one forms appearing in equation (3.1) as[1]

$$\bar{\not{\partial}}_\mu \equiv \mathrm{e}_\mu\, \partial_{\bar{q}}\, \bar{\mathrm{e}}_\mu, \quad \bar{\chi}_\mu \equiv \mathrm{e}_\mu\, \mathrm{d}\bar{q}\ \bar{\mathrm{e}}_\mu. \tag{3.2}$$

[1]There is some ambiguity in these definitions, for example we could choose $\bar{\not{\partial}}_\mu = \bar{\partial}_q\, \mathrm{e}_\mu$ and $\bar{\chi}_\mu \equiv \bar{\mathrm{e}}_\mu\, \mathrm{d}\bar{q}$, which simplifies the discussion of how rotations act on these operators. The definitions used here, however, provide significant simplifications when discussing higher degree forms, so we will use those conventions.

These operators satisfy the relations

$$\chi_\mu \left(\boxslash_\nu\right) = \delta_{\mu\nu}, \quad \sum_\mu \boxslash_\mu \boxslash_\mu = \Box, \quad 4\,\bar{\boxslash}_\mu \boxslash_\mu = \Delta,$$

where $\Box$ represents the D'Alembert operator, and the first operation represents a contraction of the one form χ_μ with the vector field $\boxslash_\mu$. We will also refer to a similar differential operator acting on the right, which will be denoted $\boxbslash_\mu$. Using this notation, a different version of (3.1) becomes

$$df = \sum_\mu \chi_\mu \, \boxslash_\mu \, f.$$

There is an analogous expression for the right $\mathbb{H}$ action derived in appendix A. This formula also readily extends to quaternion valued forms, in which case the operator acts as an (anti-)derivation. For example we have

$$\mathrm{d}(\omega \wedge \nu) = (-1)^p \sum_\mu \left(\omega \, \boxbslash_\mu \wedge \chi_\mu \wedge \nu + \omega \wedge \chi_\mu \wedge \boxslash_\mu \nu\right),$$

where ω is a p-form, with many possible variations from equations (A.4) and (A.5). It is also natural to define the conjugate Hodge-star operator, as used in complex differential geometry, which will be denoted $\bar{\star}$. Some useful formulas using this operator, proved in appendix B, are

$$\chi_\mu \wedge \bar{\star}\chi_\nu = -\bar{\star}\chi_\nu \wedge \chi_\mu = \delta_{\mu\nu}\, v, \quad \omega \wedge \bar{\star}\,\omega = <\omega, \omega> v, \tag{3.3}$$

where $<\omega, \omega>$ denotes the Hermitian inner product between forms, and v is the volume form. These formulas will be important when considering analogs of Cauchy's integral theorem.

To help interpret these operators and one forms, we can write a quaternion as an element of $\mathbb{C}^2$,

$$q = v + w\,\mathrm{e}_j \;:\; v = t + x\,\mathrm{e}_i, \quad w = y + z\,\mathrm{e}_i.$$

From twistor theory we know that this decomposition corresponds to a choice of embedding $\mathbb{H} \to \mathbb{C}^2$ that can be parameterized by a unit quaternion, which is the fiber of the mapping $P_3(\mathbb{C}) \to P_1(\mathbb{H})$ [3]

$$[v_1, w_1, v_2, w_2] \to [v_1 + w_1 j, v_2 + w_2 j],$$

corresponding to a choice of e_i. With this embedding, the $\boxslash$ operators become

$$\boxslash_0 = \partial_v - \mathrm{e}_j \partial_w, \quad \boxslash_i = \partial_v + \mathrm{e}_j \partial_w, \quad \boxslash_j = \partial_{\bar{v}} - \mathrm{e}_j \partial_{\bar{w}}, \quad \boxslash_k = \partial_{\bar{v}} + \mathrm{e}_j \partial_{\bar{w}}, \tag{3.4}$$

and the one-forms are similarly written

$$\chi_0 = \mathrm{d}v + \mathrm{d}w\,\mathrm{e}_j, \quad \chi_1 = \mathrm{d}v - \mathrm{d}w\,\mathrm{e}_j, \quad \chi_2 = \mathrm{d}\bar{v} + \mathrm{d}\bar{w}\,\mathrm{e}_j, \quad \chi_3 = \mathrm{d}\bar{v} - \mathrm{d}\bar{w}\,\mathrm{e}_j. \tag{3.5}$$

We can also investigate how these operators transform under rotation. For any proper rotation $\mathrm{d}x^R = R^T \mathrm{d}x$, there exist unit quaternions a and b such that $\mathrm{d}q^R = a\,\mathrm{d}q\,b$ and $\partial_q^R = \bar{b}\,\partial_q \bar{a}$. Under a reverse transformation, $\mathrm{e}'_\mu = \bar{a}\,e_\mu \bar{b}$, the one-forms transform as $\chi_\mu'^R = (a e'_\mu b)\,\mathrm{d}q'^R\,(\bar{b}\,\bar{\mathrm{e}}'_\mu \bar{a})$, and similarly for $\boxslash_\mu$. In addition, summed combinations of the form $\mathrm{d} = \sum_\mu \chi_\mu \boxslash_\mu$ are rotationally invariant.

4 Quaternion integral formulas

It was shown by Fueter and Sudbery [8][9] that left regular functions, defined by $\bar{\not\partial}_0 f = 0$, satisfy an integral formula given by[2]

$$f(q_0) = \frac{1}{\pi^2} \int_{\partial\Omega} \frac{(q-q_0)^{-1}}{|q-q_0|^2} \, Dq \, f(q) \tag{4.1}$$

for $q_0 \in \Omega$. Here Dq is a quaternion valued three form, which turns out to be equal to $\bar{\star}\, \mathrm{d}\bar{q}$. Using the relations in (3.3), equation (4.1) is readily proved by using Stokes' theorem to find

$$\begin{aligned} &\int_{\partial\Omega} \frac{(q-q_0)^{-1}}{|q-q_0|^2} \, \bar{\star}\, \mathrm{d}\bar{q} \, f(q) = \\ &\sum_\mu \int_\Omega \left[k(q-q_0) \boxslash_0 \bar{\star}\bar{\chi}_0 \wedge \bar{\chi}_\mu \bar{\not\partial}_\mu f(q) - k(q-q_0) \boxslash_0 \bar{\not\partial}_\mu \, \bar{\chi}_\mu \wedge \bar{\star}\bar{\chi}_0 \, f(q) \right] = \\ &\qquad\qquad - \frac{1}{4} \int_\Omega \Delta \, k(q-q_0) \, \bar{\chi}_0 \wedge \bar{\star}\bar{\chi}_0 \, f(q) = \pi^2 f(q_0), \end{aligned} \tag{4.2}$$

where we have defined $k(q) = |q|^{-2}$. We can also define the general operator $\bar{\not\partial}_\eta = \eta \, \bar{\partial}_q \, \bar{\eta}$ for a unit quaternion η, with the corresponding one form, and consider functions f^η satisfying $\bar{\not\partial}_\eta f^\eta = 0$. A similar argument to (4.2) shows that these functions satisfy the integral formula

$$\pi^2 f^\eta(q_0) = \int_{\partial\Omega} k(q-q_0) \boxslash_\eta \bar{\star}\bar{\chi}_\eta \, f^\eta(q), \tag{4.3}$$

which suggests that f^η may be more naturally interpreted as a differential form. This point of view will be developed further.

To understand the structure of the differential forms, it is important to determine how antisymmetry of the wedge product translates to the basis χ_μ. To do this, it is useful to use the complex embedding defined in (3.5), while keeping e_i and e_j arbitrary until the end of the calculation. Using this embedding we find the equalities

$$\sum_\mu \chi_\mu \wedge \bar{\chi}_\mu = 0, \quad \chi_\mu \wedge \bar{\chi}_\nu = \chi_\nu \wedge \bar{\chi}_\mu, \quad \chi_0 \wedge \bar{\chi}_i = -\chi_j \wedge \bar{\chi}_k,$$

with similar equations for the opposite ordering of χ and $\bar{\chi}$, which is related by a change of orientation. These equations are not nearly as simple if one considers expressions such as $\chi_\mu \wedge \chi_\nu$, probably because of their transformation under rotations. Additionally, we can interpret this basis for the two forms readily in terms of self-duality. In particular, we define the self-dual two-forms

$$\begin{aligned} \vartheta_i^+ &= \mathrm{e}_i \, \bar{\chi}_0 \wedge \chi_i = -\frac{1}{2}(\mathrm{d}x_0 \wedge \mathrm{d}x_i + \mathrm{d}x_j \wedge \mathrm{d}x_k), \\ \omega_\mu^+ &= \chi_\mu \wedge \bar{\chi}_\mu = \mathrm{e}_\mu \sum_i \mathrm{e}_i \, \vartheta_i^+ \, \bar{\mathrm{e}}_\mu, \end{aligned}$$

with a similar definition for the anti-self-dual forms (see appendix B). Some useful relations are

$$\omega_\mu^+ \wedge \omega_\nu^+ = -\frac{1}{2}(4\,\delta_{\mu\nu} - 1)\, v, \quad \omega_\mu^+ \wedge \chi_\nu = \frac{1}{2}\,(4\,\delta_{\mu\nu} - 1)\, \bar{\star}\bar{\chi}_\nu. \tag{4.4}$$

These equations provide all the tools necessary to evaluate three-forms and four-forms in terms of the basis we've defined. In appendix B we use this method to write the de Rham complex with this basis.

[2]Here Dq is half of the value in [9] to be consistent with our conventions.

It is now possible to interpret (4.3) as an integral formula for differential forms. This is accomplished by using (4.4) to replace $\bar{\star}\bar{\chi}_\eta$ in equation (4.3), and rearranging terms. For example, if we consider a one-form $f^{(1)} = \sum_\mu \chi_\mu f_\mu^{(1)}$ satisfying $\bar{\not{\partial}}_\mu f_\mu^{(1)} = 0$, we find

$$\int_{\partial\Omega} \sum_\mu \Big(\chi_\mu(p) \otimes k(p-q) \,\bar{\not{\times}}_{\mu,q}\, \omega_\mu^+(q)\Big) \wedge f^{(1)}(q) = 2\pi^2 \Big(f^{(1)}(p) - \frac{1}{4}\sum_\mu \chi_\mu(p) \sum_\nu f_\nu^{(1)}(p)\Big).$$

The second term on the right prevents this from being a useful integral formula, unless we demand $\sum_\mu f_\mu^{(1)}(p) = 0$, which would make the term vanish. A different approach is to consider self-dual two forms, for which this condition comes from the fact that the ω_μ^+ form an over-complete basis. Indeed, for a self dual two-form $f^{(2)} = \sum_\mu \omega_\mu^+ f_\mu^{(2)}$ satisfying $\not{\partial}_\mu f_\mu^{(2)} = 0$, we can choose the coefficients $f_\mu^{(2)}$ to satisfy $\sum_\mu f_\mu^{(2)} = 0$, and we find

$$f^{(2)}(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} G^+(p,q) \wedge f^{(2)}(q), \tag{4.5}$$

where we have defined

$$G^+(p,q) = \sum_\mu \omega_\mu^+(p)\, \bar{\not{\partial}}_\mu k(p-q) \otimes \bar{\chi}_\mu(q). \tag{4.6}$$

From (B.1) we find that this is in fact an integral representation for a closed self-dual two-form. This formula leads directly to a generalization of equation (2.3) to the quaternions. To see this, we define the kernel taking values in $T^*\Omega \otimes T^*\Omega$

$$K(p,q) = \sum_\mu \frac{\chi_\mu(p) \otimes \bar{\chi}_\mu(q)}{|p-q|^2} = \sum_\mu \frac{\bar{\chi}_\mu(p) \otimes \chi_\mu(q)}{|p-q|^2} = 4\,\mathrm{Re}\left(\frac{\mathrm{d}p \otimes \mathrm{d}\bar{q}}{|p-q|^2}\right), \tag{4.7}$$

which satisfies the differential equation

$$-\,\mathrm{d}_p K(p,q) = G^+(p,q) + G^-(p,q),$$

where G^- is related to G^+ by a change of orientation. To prove this, we can insert d in the form $\sum_\mu \chi_\mu \not{\partial}_\mu$ into the first expression for K(p,q), and $\sum_\mu \bar{\chi}_\mu \bar{\not{\partial}}_\mu$ into the second expression. We then choose only the self-dual terms, and use the second form of the expression to get the anti-self-dual terms.

We use a similar method to find the second differential of K(p,q), as outlined in appendix C,

$$\begin{aligned}\mathrm{d}_p\mathrm{d}_q K(p,q) = \sum_i \Big[& 4\pi^2 \delta(p-q)\Big(\vartheta_i^+ \otimes \vartheta_i^+ \,+\, \vartheta_i^- \otimes \vartheta_i^-\Big) \\ & + \sum_{j,k} \boxtimes_{jk} k(p-q)\Big(\vartheta_j^- \otimes \vartheta_k^+ + \vartheta_k^+ \otimes \vartheta_j^-\Big)\Big],\end{aligned} \tag{4.8}$$

where we have defined the second order differential operators

$$\boxtimes_{jk} k(q) = 4\,\mathrm{Re}\Big(\partial_{\bar{q}}\, \mathrm{e}_j\, \partial_q\, \bar{\mathrm{e}}_k\, k(q)\Big) = \begin{cases} (\partial_0^2 + \partial_i^2 - \partial_j^2 - \partial_k^2)\, k(q) & \text{if } j = k, \\ (\epsilon_{ijk}\, \partial_{\{0,}\partial_{i\}} + \partial_{\{j,}\partial_{k\}})k(q) & \text{if } j \neq k. \end{cases} \tag{4.9}$$

These expressions lead to a generalization of (2.1) for the quaternions. For two closed self-dual two-forms f and g, we have

$$\int_\Omega f(p) \wedge g(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} \int_{\partial\Omega} f(p) \wedge K(p,q) \wedge g(q),$$

In addition, if an arbitrary two form satisfies both $\mathrm{d}F = 0$ and $\mathrm{d}\bar{\star}F = 0$, then we obtain

$$\int_\Omega F(p) \wedge \bar{F}(p) = \frac{1}{2\pi^2} \int_{\partial\Omega} \int_{\partial\Omega} \mathrm{Re}\Big(F(p) \wedge K(p,q) \wedge \bar{\star} F(q) \Big) \tag{4.10}$$

This expression looks suggestively similar to a topological invariant that arises in the study of Yang-Mills theory, known in physics as the instanton number. This number represents an absolute lower bound for the energy of a field configuration in a given homotopy class, which is saturated by (anti-)self-dual connections known as instantons [3][14]. The conditions on F are also analogous to the Bianchi identity, and the Yang-Mills equation of motion. Our expression, however, is not currently relevant to Yang-Mills theory, because we are dealing with the exterior derivative instead of a gauge-covariant derivative. Before attempting to generalize to this case, however, we will define a modified integral kernel that has better properties than K(p,q).

5 Modified integral kernel

The cross terms in equation (4.8) restrict the usage of this integral kernel in more general formulae, so we will seek a new kernel $\tilde{K}$ that does not have these cross terms. We will be particularly interested in an analogue of (4.10) for the Lagrangian of Yang-Mills theory

$$\mathcal{L} = -\int_\Omega F(p) \wedge \bar{\star} F(p). \tag{5.1}$$

For now we require that F(p) is a closed two-form, but this will eventually be replaced with the Bianchi identity on the curvature F,

$$D_A F = \mathrm{d}F + A \wedge F - F \wedge A = 0.$$

To motivate our definition of $\tilde{K}$, we introduce the quaternion regular function L(q) defined in [9], which is a primitive for the kernel appearing in (4.1),

$$2\,L(q) = \begin{cases} (q\,\mathrm{Im}\,q)^{-1} + \left| \dfrac{\arg(q)}{(\mathrm{Im}\,q)^2} \right| i_q & \text{if } \mathrm{Im}\,q \neq 0, \\ -\dfrac{1}{|q|^2} & \text{if } \mathrm{Im}\,q = 0. \end{cases} \tag{5.2}$$

Here Im(q) denotes the imaginary part of q, i_q is the imaginary unit $\frac{\mathrm{Im}(q)}{|\mathrm{Im}(q)|}$, and arg(q) is the argument in the complex plane generated by i_q. L(q) satisfies the relations

$$\partial_{\bar{q}} L(q) = 0, \quad \partial_q L(q) = \frac{q^{-1}}{|q|^2}, \tag{5.3}$$

which will be useful for defining a modified integral kernel by isolating the cross terms in equation (4.8). The branch cut introduced by the complex argument requires some care to treat properly, requiring methods from complex analysis and distribution theory.

We start by defining a kernel using $L_{\tilde{\mu}}(p-q) = \mathrm{e}_\mu L(p-q)\bar{\mathrm{e}}_\mu$

$$K^{\pm}(p,q) = \sum_\mu \chi_\mu(p) \otimes L_{\tilde{\mu}}(p-q)\, \bar{\chi}_\mu(q) = 4\,\mathrm{Re}\left(\mathrm{d}p \otimes L(p-q)\,\mathrm{d}\bar{q}\right). \tag{5.4}$$

Fixing the value of p, we see that L(p,q) is singular for $q \in S^p = \{q \in \Omega : \mathrm{Re}(p-q) \leq 0, \mathrm{Im}(p-q) = 0\}$. In this region the behavior of the differential is not readily apparent, and we will consider an analogue

of the "keyhole" contour to make it well-defined, illustrated in figure 1. We define a neighborhood of S^p by $S^p_{\epsilon,r} = U^p_\epsilon \cup C^p_r$ where C^p_r is a cylinder of radius r around S^p, and U^p_ϵ is a ball of radius ϵ around q=p. We also define the complement $\Omega^p_{\epsilon,r} = \Omega \backslash S^p_{\epsilon,r}$. Considering a point $q \in \Omega^p_{\epsilon,r}$, we find

$$\mathrm{d}_p \mathrm{d}_q K^{\pm} = -\sum_{j,k} \boxtimes_{jk} k(p-q)\, \vartheta^-_j \otimes \vartheta^+_k .$$

We can also define the related kernel $K^{\mp}(p,q) = \sum_\mu \bar{\chi}_\mu(p) \otimes \bar{L}_{\tilde{\mu}}(p-q)\, \chi_\mu(q)$, satisfying the similar equation $\mathrm{d}_p \mathrm{d}_q K^{\mp} = -\sum_{j,k} \boxtimes_{jk} k(p-q)\, \vartheta^+_k \otimes \vartheta^-_j$. These relations motivate the definition

$$\tilde{K}(p,q) = K(p,q) + K^{\pm}(p,q) + K^{\mp}(p,q), \tag{5.5}$$

that satisfies $\mathrm{d}_p \mathrm{d}_q \tilde{K}(p,q) = 0$ for $q \in \Omega^p_{\epsilon,r}$.

The fact that the cross terms cancel outside of the singular region is encouraging, but there is still a contribution from S^p. To understand the behavior in this region, we first consider the behavior of L(p,q) in the patch $P^p_r = \partial\Omega \cap S^p_{\epsilon,r}$. In this patch L(p,q) has a form similar to the Riesz kernel, and its integral diverges logarithmically. To make the integral over $\partial\Omega$ well defined, we will take something similar to the Cauchy principal value, integrating only over $\partial\Omega^p_r = \partial\Omega \backslash P^p_r$. The keyhole contour will be denoted $\Gamma^p_{\epsilon,r}$, and we write

$$-\int_{\partial\Omega^p_r} \mathrm{d}_p \tilde{K}(p,q) \wedge F(q) = \int_{\Omega^p_{\epsilon,r}} \mathrm{d}_p \tilde{K}(p,q) \wedge \mathrm{d}_q F(q) + \int_{\Gamma^p_{\epsilon,r}} \mathrm{d}_p \tilde{K}(p,q) \wedge F(q). \tag{5.6}$$

We will compute the integral over $\Gamma^p_{\epsilon,r}$, considering first the cylinder as $r \to 0$, and then the sphere with $\epsilon \to 0$. On the cylinder, K(p,q) is not divergent, so it does not contribute as r goes to zero. To find the contribution from $K^{\pm}$, we fix a point $q \in S^p$, and Taylor expand F(q) to first order for a small imaginary perturbation $\mathbf{r}$, $F(q+\mathbf{r}) = F(q) + \sum_i \partial_i F(q) r_i + O(r^2)$. In appendix C we show that the contributions of order r^2 vanish as $r \to 0$, in addition to the contribution from the constant term. For the differential, we calculate

$$\mathrm{d}_p K^{\pm}(p-q) = -4\,\mathrm{Re}\Big(\partial_{\bar{q}} \omega_0^- L(q-p) \mathrm{d}\bar{q}\Big). \tag{5.7}$$

Denoting the surface of the cylinder by $\Gamma^p_{C,r}$, we perform the two-dimensional integral over $\mathbf{r}$ with fixed radius, which yields

$$\lim_{r\to 0} \Big(\int_{\Gamma^p_{C,r}} \mathrm{d}_p K^{\pm}(p.q) \wedge F(q) \Big) = \frac{2\pi^2}{3} \sum_i \vartheta^-_i(p) \int_{S^p} \Big(\partial_t (F^+_i(q) + F^-_i(q)) - 2\,\mathrm{d}F(q)\rfloor_i \Big)\, \mathrm{d}t,$$

where t denotes the real part of q, $\mathrm{d}F(q)\rfloor_i$ denotes the projection of $\mathrm{d}F(q)$ onto $\star \mathrm{d}x_i$, and $F^{\pm}_i(q)$ denotes the projection of $F(q)$ onto $\vartheta^{\pm}_i$. Some of the details of this calculation are shown in appendix C.

We now consider the integral over the sphere ∂U^p_ϵ. The differential of K(p,q) is given by

$$\mathrm{d}_p K(p-q) = 4\mathrm{Re}\Big(\omega_0^+ \partial_{\bar{q}} k(q-p) \mathrm{d}\bar{q} - \partial_{\bar{q}} k(q-p) \omega_0^- \mathrm{d}\bar{q} \Big).$$

Using this expression, in addition to equation (5.7), we find (CHECK SIGNS)

$$\lim_{\epsilon\to 0} \Big(\int_{\partial U^p_\epsilon} \mathrm{d}_p \tilde{K}(p.q) \wedge F(q) \Big) = 4\pi^2 \star F(p) + \frac{2\pi^2}{3} \varrho\, F(p),$$

where ϱ is an orientation-reversing operator that acts on two forms through $\varrho\, \vartheta^{\pm}_i = \star \vartheta^{\mp}_i$.

A Conventions and Formulas

In this paper, we will not use the Einstein summation convention, because there will frequently be repeated indices that are not summed over. This is due to some of the definitions that are used, which simplify calculations overall. In addition, whenever i, j, and k appear as subscripts without summation, it will be assumed that $\mathrm{e}_i \mathrm{e}_j = \epsilon_{ijk} \mathrm{e}_k$.

For complex numbers, we will use the following convention to avoid factors of 2:

$$\mathrm{d}z = \mathrm{d}x + i\mathrm{d}y, \quad \partial_z = \frac{1}{2}(\partial_x - i\partial_y).$$

The Dolbeault operators are defined in the usual way, with the conventions following [15].

For the quaternions we will make a similar choice, defining

$$\mathrm{d}q = \frac{1}{2}\sum_\mu (\mathrm{e}_\mu \mathrm{d}x_\mu), \quad \partial_q = \frac{1}{2}\sum_\mu (\bar{\mathrm{e}}_\mu \partial_\mu).$$

We will also sometimes use the notation $\{t, x_i\}$ for the components of q.

The following formulas are used extensively throughout the text,

$$\mathrm{Re}(q) = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \, q \, \mathrm{e}_\mu, \quad \mathrm{Re}(pq) = \mathrm{Re}(qp). \tag{A.1}$$

Another set of useful relations, closely related to (A.1), are

$$q_\nu = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \, q \, \bar{\mathrm{e}}_\nu \mathrm{e}_\mu = \frac{1}{4}\sum_\mu \bar{\mathrm{e}}_\mu \bar{\mathrm{e}}_\nu \, q \, \mathrm{e}_\mu, \tag{A.2}$$

$$\bar{q} = -\frac{1}{2}\sum_\mu e_\mu \, q \, e_\mu. \tag{A.3}$$

We note that the the exterior derivative operator, $\mathrm{d} = \sum_\mu \mathrm{d}x_\mu \partial_\mu$, can be written $4\,\mathrm{Re}(\mathrm{d}q\,\partial_q)$, so using the definitions in equation (3.2), we write

$$\mathrm{d}f = \sum_\mu \chi_\mu (\boxslash_\mu f) = \sum_\mu (f \boxbslash_\mu)\, \chi_\mu, \tag{A.4}$$

$$\mathrm{d}f = \sum_\mu \bar{\chi}_\mu (\bar{\boxslash}_\mu f) = \sum_\mu (f \bar{\boxbslash}_\mu)\, \bar{\chi}_\mu. \tag{A.5}$$

Because all of these formulas come from writing the real (anti-commuting) exterior derivative with quaternions, we can immediately derive analogues of the product rule, such as

$$\mathrm{d}(\omega \wedge \nu) = (-1)^p \sum_\mu \left(\omega \, \bar{\boxbslash}_\mu \wedge \bar{\chi}_\mu \wedge \nu + \omega \wedge \chi_\mu \wedge \boxslash_\mu \nu \right).$$

It is also possible to write an arbitrary quaternion-valued form using a similar decomposition. In this case, we write $A = \sum_\mu a_\mu \mathrm{d}x_\mu$, and use equation (A.2) to write $\mathrm{d}x_\mu$ in terms of $\mathrm{d}q$, then rearrange terms to find $A = \sum_\mu \alpha_\mu \chi_\mu$, where $\alpha_\mu = \frac{1}{2}\sum_\nu (a_\mu \, \mathrm{e}_\mu \bar{\mathrm{e}}_\nu \bar{\mathrm{e}}_\mu)$. Conversely, we find that $a_\mu = \frac{1}{2}\sum_\nu (\alpha_\mu \, \mathrm{e}_\mu \mathrm{e}_\nu \bar{\mathrm{e}}_\mu)$. A similar procedure can be used for any degree form with any complete basis.

B Quaternion valued forms

A quaternion valued one-form is an element $\omega \in \mathbb{H} \otimes T^*M$ that acts on a quaternion valued vector $V \in \mathbb{H} \otimes TM$ by contraction and multiplication. The conjugate star operator acts naturally on quaternion valued one-forms when they are written using the χ_μ basis. In particular, the χ_μ basis satisfies

$$\chi_\mu \wedge \bar{\star}\chi_\nu = -\bar{\star}\chi_\nu \wedge \chi_\mu = \delta_{\mu\nu}\, v.$$

To prove this equation, we write $\chi_\mu \wedge \bar{\star}\chi_\nu = \mathrm{e}_\mu \mathrm{d}q\, (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \wedge (\bar{\star}\mathrm{d}q)\, \bar{\mathrm{e}}_\nu$. Noting that only the middle terms are relevant, we calculate

$$\mathrm{d}q\, (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \wedge \bar{\star}\mathrm{d}q = \frac{1}{24} \sum_{\alpha,\sigma} \mathrm{e}_\alpha (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\sigma\, \epsilon_{\sigma\gamma\delta\rho}\, \mathrm{d}x^\alpha \wedge \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta \wedge \mathrm{d}x^\rho =$$
$$\frac{1}{4} \sum_{\alpha,\sigma} \mathrm{e}_\mu (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\sigma\, \delta_{\alpha\sigma}\, v = \frac{1}{4}\, v \sum_\alpha \mathrm{e}_\alpha (\mathrm{e}_\mu \bar{\mathrm{e}}_\nu) \bar{\mathrm{e}}_\alpha = v\, \mathrm{Re}(e_\mu \bar{\mathrm{e}}_\nu) = v\, \delta_{\mu\nu}.$$

The second equality is proved in a similar way. This provides two natural bases for the three forms, given by $\bar{\star}\chi_\mu$ and $\bar{\star}\bar{\chi}_\mu$. The two forms are decomposed by direct calculation as described in the text, and using the techniques outlined we can write down the full de Rham complex in our basis:

$$\begin{aligned}
\mathrm{d} f^{(0)} &= \sum_\mu \chi_\mu \boxtimes_\mu f^{(0)}, \\
\mathrm{d} \sum_\mu \chi_\mu\, f^{(1)}_\mu &= -\sum_\mu \omega^+_\mu\, \bar{\boxtimes}_\mu f^{(1)}_\mu - \frac{1}{2} \sum_i \omega^-_i \left(\bar{\boxtimes}_{\{0} f^{(1)}_{i\}} - \bar{\boxtimes}_{\{j} f^{(1)}_{k\}} \right), \\
\mathrm{d} \sum_\mu \left(\omega^+_\mu\, f^{(2)}_\mu + \omega^-_\mu\, g^{(2)}_\mu \right) &= 2 \sum_\mu \left(\bar{\star}\bar{\chi}_\mu \boxtimes_\mu\, f^{(2)}_\mu - \bar{\star}\chi_\mu \bar{\boxtimes}_\mu g^{(2)}_\mu \right), \\
\mathrm{d} \sum_\mu \bar{\star}\chi_\mu f^{(3)}_\mu &= \sum_\mu \left(\boxtimes_\mu\, f^{(3)}_\mu \right) v,
\end{aligned} \tag{B.1}$$

where we have chosen $f^{(2)}_\mu$ to satisfy $\sum_\mu f^{(2)}_\mu = 0$, and a similar relation for $g^{(2)}$. Here, $\{\}$ represents symmetrization of indices.

It is also useful to have analogues of (4.4) for anti-self dual forms. We get

$$\omega^-_\mu \wedge \omega^-_\nu = \frac{1}{2} (4\delta_{\mu\nu} - 1)\, v, \quad \omega^-_\mu \wedge \bar{\chi}_\nu = -\frac{1}{2} (4\delta_{\mu\nu} - 1)\, \bar{\star}\chi_\nu.$$

The second identity is proved by writing the right hand side as an arbitrary linear combination of three forms, $\sum_\nu a_\nu \bar{\star}\chi_\nu$, then taking the wedge product of both sides with χ_ρ and comparing coefficients. A similar calculation shows that

$$\vartheta^+_i \wedge \bar{\chi}_\mu = \frac{\mathrm{e}_i}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\chi_{\mu_i}, \quad \vartheta^-_i \wedge \chi_\mu = -\frac{\mathrm{e}_i}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\chi_{\mu_i}, \tag{B.2}$$

where we have defined μ_i so that $\bar{\mathrm{e}}_\mu \mathrm{e}_{\mu_i} = \mathrm{qs}(\mu_i)\mathrm{e}_i$, and

$$\mathrm{hs}(\mu_i) = \begin{cases} 1 & \text{if } \mu = 0, i \\ -1 & \text{if } \mu = j, k, \end{cases} \qquad \mathrm{qs}(\mu_i) = \begin{cases} 1 & \text{if } \mu = 0, k \\ -1 & \text{if } \mu = i, j. \end{cases}$$

A useful identity with these definitions is $\bar{\mathrm{e}}_i \mathrm{e}_{\mu_i} = \mathrm{hs}(\mu_i)\mathrm{qs}(\mu_i)\mathrm{e}_\mu$. Also, taking the complex conjugate of both sides of (B.2) yields similar expressions such as

$$\chi_\mu \wedge \vartheta^+_i = -\frac{1}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\bar{\chi}_{\mu_i} \mathrm{e}_i, \quad \bar{\chi}_\mu \wedge \vartheta^-_i = \frac{1}{2}\, \mathrm{hs}(\mu_i)\, \bar{\star}\chi_{\mu_i} \mathrm{e}_i, \tag{B.3}$$

$$\bar{\chi}_\nu \wedge \omega^+_\mu = -\frac{1}{2} (4\delta_{\mu\nu} - 1) \bar{\star}\chi_\nu, \quad \chi_\nu \wedge \omega^-_\mu = \frac{1}{2} (4\delta_{\mu\nu} - 1) \bar{\star}\bar{\chi}_\nu. \tag{B.4}$$

C Useful equations for integral kernels

The integral kernel K(p,q) was defined in equation (4.7), leading to expression (4.8) for the second differential. This expression is most easily obtained by isolating the self-dual and anti-self-dual terms. For example, for the terms coupling self-dual to anti-self dual forms we find

$$\begin{aligned}\Big(\mathrm{d}_p\mathrm{d}_q K(p-q)\Big)_{\pm} &= \sum_{\mu,\,\nu\neq\mu} \chi_\mu(p)\wedge\bar\chi_\mu(p)\,\bar\boxdot_\mu\, k(p-q)\,\boxslash_\nu\,\chi_\nu(q)\wedge\bar\chi_\mu(q)\\ &= \sum_i \Big(\mathrm{e}_i\vartheta_i^+(p)(\bar\boxdot_{\{0,}\boxdot_{i\}}+\bar\boxdot_{\{j,}\boxdot_{k\}})\,k(p-q)\,\mathrm{e}_i\vartheta_i^-(q)+\mathrm{e}_j\vartheta_j^+(\bar\boxdot_{[0,}\,\boxdot_{k]}+\bar\boxdot_{[i,}\boxdot_{j]})\,k(p-q)\,\mathrm{e}_k\vartheta_k^-\Big)\\ &= -4\sum_{j,k}\boxtimes_{jk}\,k(p-q)\,\vartheta_j^+(p)\otimes\vartheta_k^-(q),\end{aligned}$$

where we have defined $k(q)=|q|^{-2}$. A similar method is used for the other terms in the expression.

In order to define the modified integral kernel in section 5, we defined the function L(q) in equation (5.2). Sudbery [9] shows that this function can be written in the more manageable form

$$L(q)=-\frac{1}{2|q|^2}+F(r,t)\,\mathbf{r},\qquad F(r,t)=\frac{1}{2r^2}\left(-\frac{t}{|q|^2}+\frac{1}{r}\tan^{-1}\left(\frac{r}{t}\right)\right),$$

where we use the convenient notation $t=q_0$, $\mathbf{r}=\sum_i \mathrm{e}_i q_i$, and $r=|\mathbf{r}|$. Using the derivatives

$$\partial_t F(r,t)=-\frac{1}{|q|^4},\quad r\,\partial_r F(r,t)=\frac{t}{|q|^4}-3\,F(r,t),\tag{C.1}$$

it is straightforward to prove the equations in (**??**). These equations are also useful for proving equation (**??**) in the region $\Omega_{p,\epsilon}$. To do this, we

For the opposite orientation, we similarly define

$$\mathrm{d}_p\mathrm{d}_q K^{\mp}(p,q)=4\sum_{j,k}\boxtimes^{jk}\bar L(p-q)\,\vartheta_j^+\otimes\vartheta_k^-,$$

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